Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	TimeOut
Project / Solution Name	Timed Production Planning
Name (Leader)	Muhammad Husein Ilhamsyah
Email & Phone (Leader)	Husein.ilhamsyah@gmail.com & +60 172717957
University Affiliation(s)	Sunway University
Live Prototype URL	Local prototype
Code Repository URL (public / access granted to judges)	https://github.com/Whoosayinn/Timed (isPublic)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The Next.js/React interface calls a Python FastAPI backend. Orders, recipes, stock and capacity feed deadline-based scheduling, a first-come-first-served comparison and purchase requests. SQLite stores current data, decisions and activity; CSV/Excel supports exchange.
A custom NumPy/pandas regression forecasts weekly demand and estimated 80% ranges. Stockout calculations combine demand with supplier lead times. Optional Groq, Gemini or OpenRouter adapters explain calculated facts; numeric checking and English template briefings provide fallbacks.
Flow: user -> interface -> FastAPI -> planning/forecasting -> SQLite/reports -> user. Optional facts -> LLM -> checked text. The frontend currently uses localhost; secure cloud deployment remains planned.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3; SQLite	Backend and persistence	PSF; public domain
Next.js 16.4.2; React/DOM 19.3.0	Web interface	MIT
TypeScript ^5; Tailwind ^4; ESLint ^9; @types; @tailwindcss/turbopack	Frontend types, styling and build	Apache-2.0 (TS); MIT (others)
FastAPI >=0.110; Pydantic >=2.6; openpyxl >=3.1; pytest >=8	API, validation, Excel, tests	MIT
NumPy >=1.26; pandas >=2.1; Uvicorn >=0.29; HTTPX >=0.27	Forecasting, server, HTTP	BSD-3-Clause
python-multipart >=0.0.9	Upload parsing	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Groq OpenAI-compatible API	Optional briefing/Q&A; adapter implemented	Usage and tier to confirm
Google Gemini OpenAI-compatible API	Optional alternative briefing/Q&A	Usage and tier to confirm
OpenRouter OpenAI-compatible API	Optional alternative briefing/Q&A	Usage and tier to confirm
Custom OpenAI-compatible endpoint	Optional environment-variable override	Configuration to confirm
Other external APIs	None identified in application code	Team to confirm

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Cloud hosting / storage / auth	No configured deployment identified; local Next.js and Uvicorn workflow	Provider, plan and region to confirm
SQLite local database	Dataset, purchase orders and activity log; not a cloud service	Local timed.db; TIMED_DB override

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Custom per-product ridge regression	Team repository core/forecast.py	Trained on synthetic sales; no pretrained weights	Repository Apache-2.0
llama-3.3-70b-versatile	Groq adapter	Optional pretrained API model; use unconfirmed	Provider terms and Meta Llama terms
gemini-flash-lite-latest	Google Gemini adapter	Optional pretrained API model; mutable alias	Google API terms
meta-llama/llama-3.3-70b-instruct:free	OpenRouter adapter	Optional pretrained API model; use unconfirmed	Provider and Meta Llama terms
LLM_MODEL override	User-configured endpoint	No configured model established	Confirm if used

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Production inputs	data/ CSV + capacity.json	Synthetic	Repo Apache-2.0	N
Sales history	data/forecasting/ sales_history.csv	Synthetic; train/test	Repo Apache-2.0	N
Event calendar	data/forecasting/ calendar_events.csv	Illustrative dates	Repo Apache-2.0	N
Example plan	specs/ + frontend/data/ output-example.json	Synthetic output	Repo Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex; version not recorded	Prepared these reports from project files	Moderate
Other development AI tools	[Team to confirm tools, versions and extent]	[Team to confirm]

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The repository includes automated tests for planning, inventory, forecasting, explanations and API behaviour. The sample planning comparison and forecast backtest were reproduced for these reports. Numeric checking and template fallbacks constrain optional AI briefings. The team must review the final reports, confirm development-tool use and verify every declaration before signing. Team review history and responsibility acceptance require confirmation.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
[Team to confirm]	No sponsor resource established by project files	Confirm API, credits, tools, mentoring or data used

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Core scheduling, forecasting, local storage and built-in explanations run without a sponsor API. Optional language-model providers can be replaced through configuration. Any actual sponsor resource use and its importance must be confirmed by the team.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The bundled datasets are synthetic and contain no real personal data. User uploads may include customer names, orders, prices, inventory and recipes. The backend persists datasets, purchase orders and activity logs in local SQLite. Optional AI calls send calculated facts, including customer/order details and commercial quantities; questions are also sent for Q&A. The code reads credentials from environment variables or a local .env file. Data has no implemented expiry, encryption-at-rest or retention policy. The X-Role header distinguishes owner and staff but is not secure authentication. Before real-data use, add authenticated access, HTTPS, data minimisation, controlled storage and deletion rules. Provider retention terms and permission to share business data need review.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Forecasts are fitted to synthetic data and may not generalise to real businesses. Calendar dates are approximate; missing weeks are skipped and extreme observations filtered. Estimated 80% ranges are not guarantees. The 26-week synthetic holdout produces MAPE of 6.8% for butter cookies, 5.3% for curry puffs and 9.2% for coconut biscuits. Earliest-due-date scheduling is a heuristic with simplified capacity and material assumptions. LLMs can misstate facts; the numeric filter checks number membership, not complete factual accuracy. Provider failures trigger template briefings or withheld answers. Free-text Q&A needs a configured provider, and offline explanations are English. Production risks include localhost connections, unauthenticated role headers and unrestricted retention. Mitigation priorities are real-data testing, authenticated roles, secure deployment and human approval of purchases.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Muhammad Husein Ilhamsyah — Whoosayinn

25 October 2026